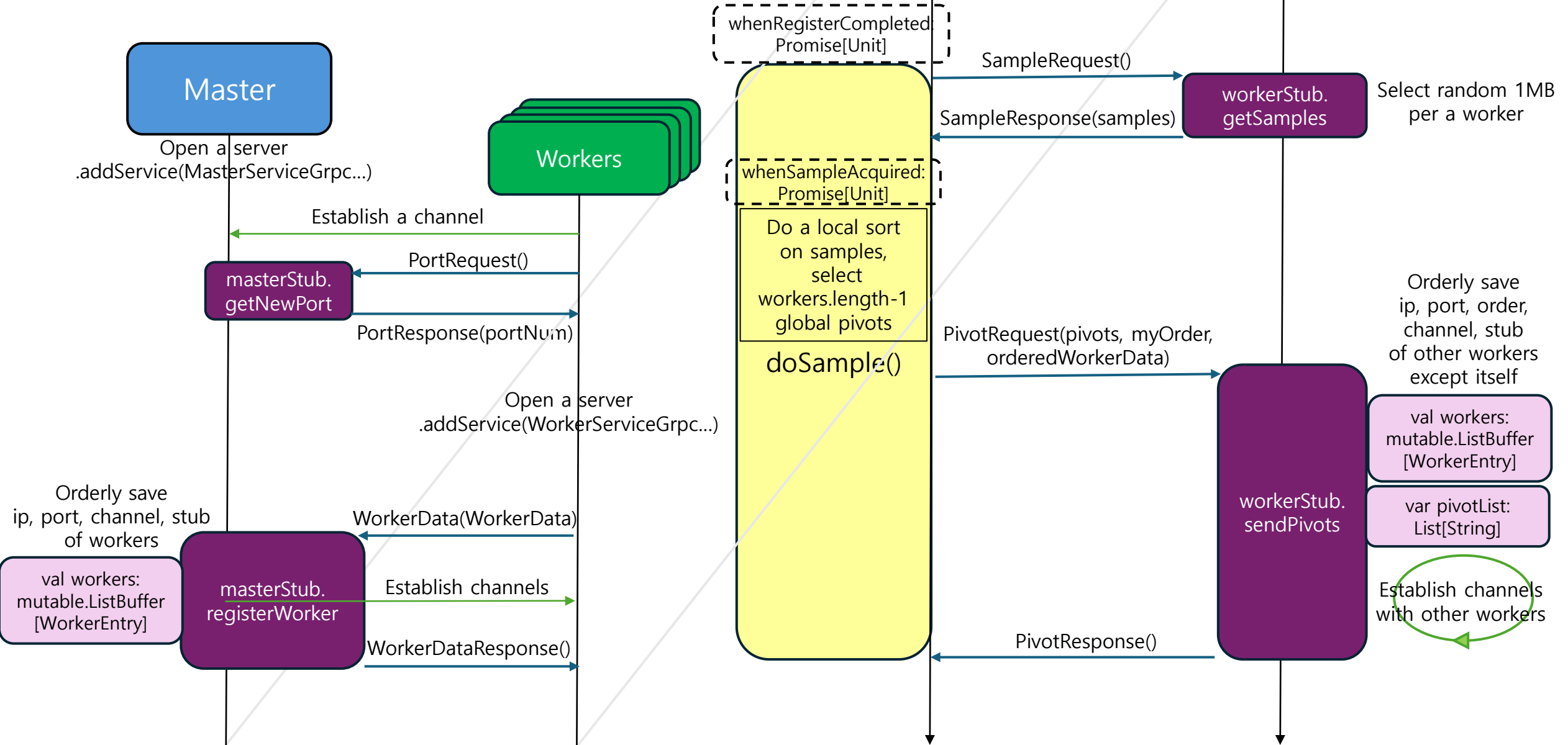


# Final Project Presentation

Team Navy

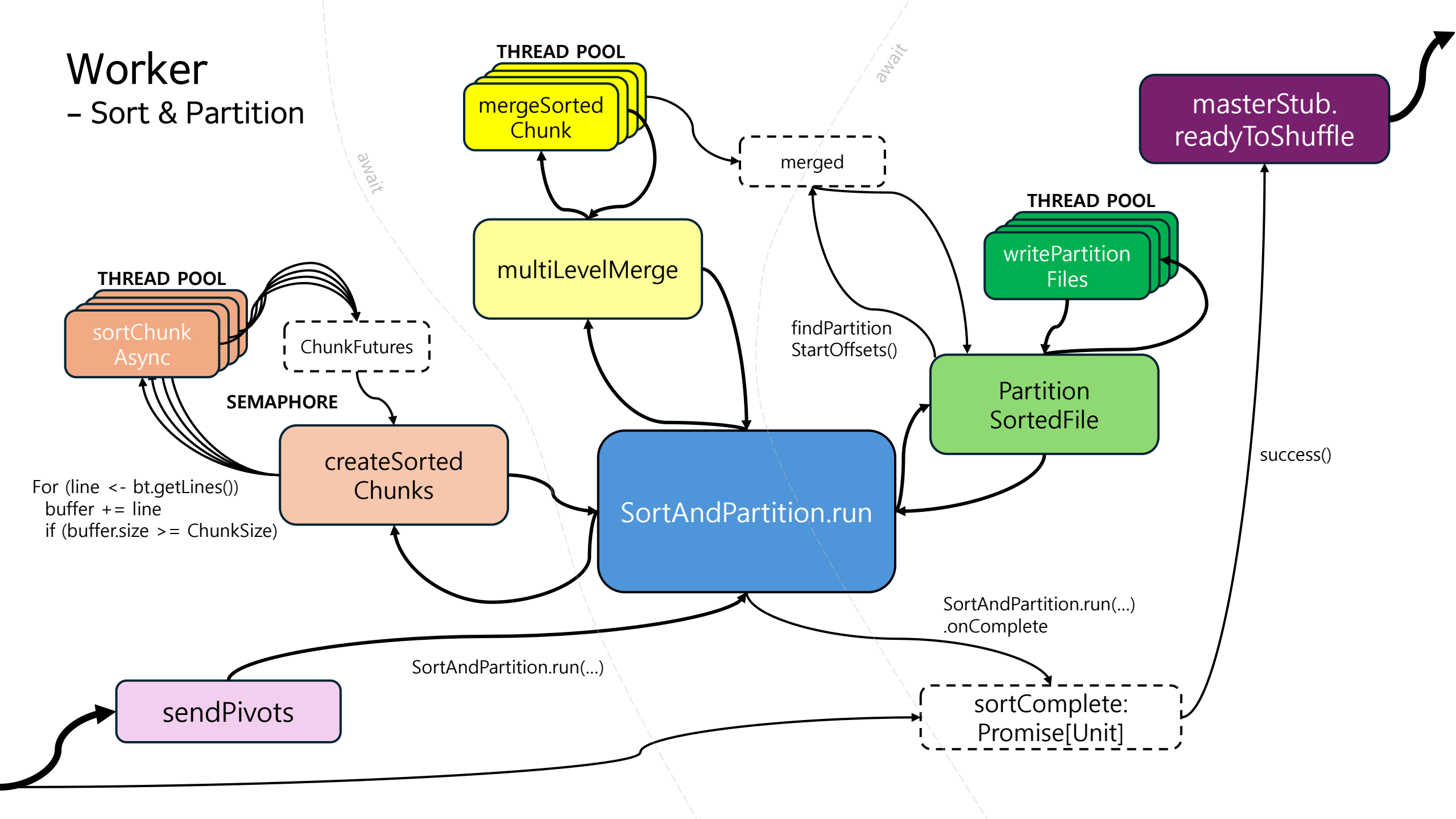
# Master

## - Initialize & Sampling

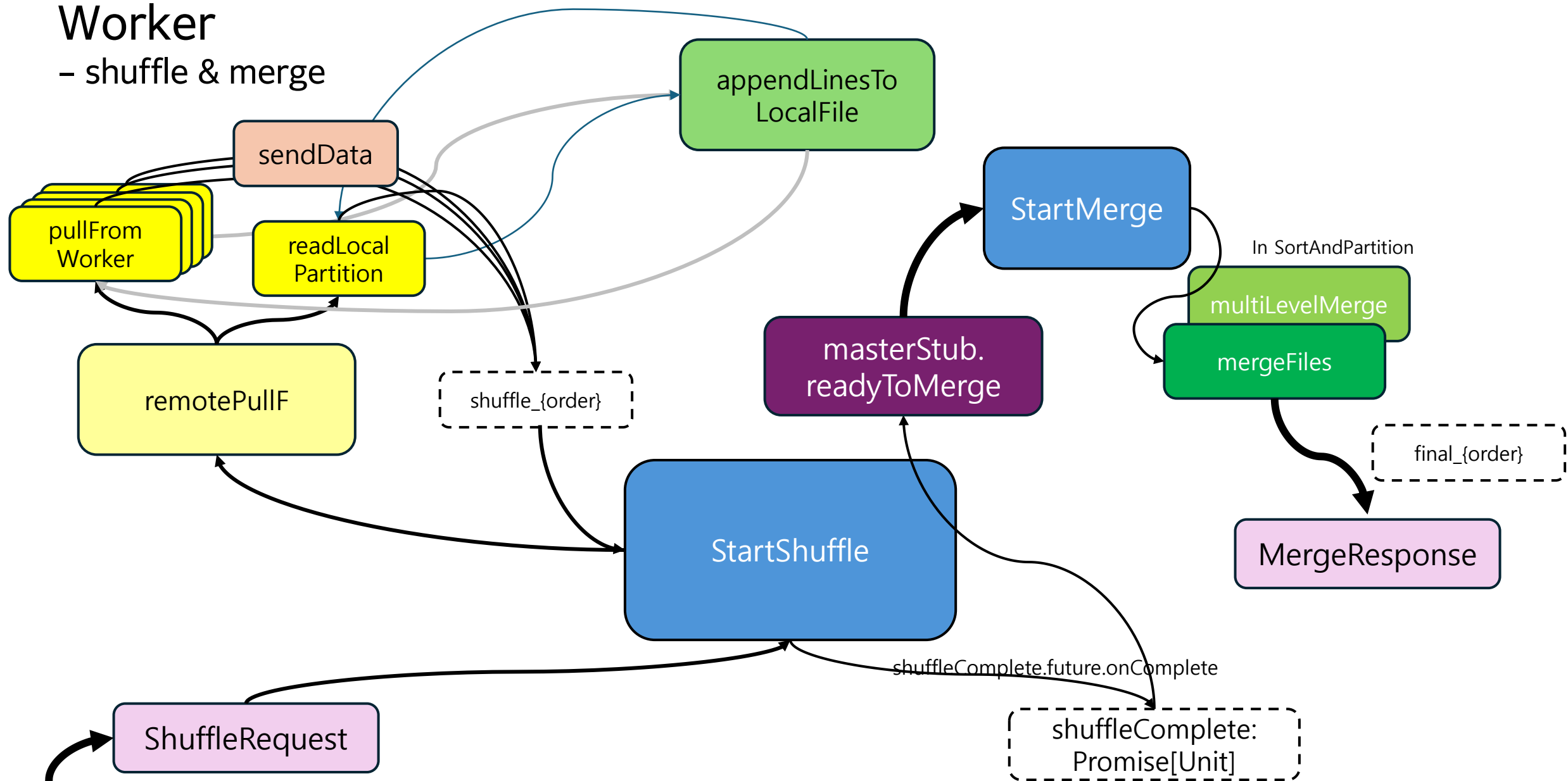


# Worker

- Sort & Partition



- shuffle & merge



# Demo

- Smalltest 10

```
navy@vm-1-master:~  
[MASTER] Worker shutdown ACK received from 2.2.2.101:50500  
[MASTER] Worker shutdown ACK received from 2.2.2.104:50504  
[MASTER] Worker shutdown ACK received from 2.2.2.102:50501  
[MASTER] All shutdown RPCs completed. Stopping master gRPC server...  
[MASTER] Master terminated.  
[MASTER] Normal shutdown in progress, skip shutdownHook server.shutdown()  
[navy@vm-1-master 332project]$  
[navy@vm-1-master 332project]$  
[navy@vm-1-master 332project]$  
[navy@vm-1-master 332project]$  
[navy@vm-1-master 332project]$  
[navy@vm-1-master 332project]$  
[navy@vm-1-master 332project]$  
[navy@vm-1-master 332project]$  
[navy@vm-1-master 332project]$  
[navy@vm-1-master 332project]$ exit  
logout  
Connection to 141.223.16.227 closed.  
  
C:\Users\leeju>ssh navy@141.223.16.227 -p 7777  
navy@141.223.16.227's password:  
Last login: Tue Dec 9 00:14:37 2025 from 172.17.114.73  
[navy@vm-1-master ~]$  
  
navy@vm-1-master:~  
sent 18,874 bytes received 27,767 bytes 31,094.00 bytes/sec  
total size is 21,332,071 speedup is 457.37  
=== Deploy to 2.2.2.104 ===  
sending incremental file list  
worker.jar  
  
sent 29,949 bytes received 27,767 bytes 2,959.79 bytes/sec  
total size is 21,332,071 speedup is 369.60  
=== Deploy to 2.2.2.105 ===  
sending incremental file list  
worker.jar  
  
sent 43,639 bytes received 27,767 bytes 47,604.00 bytes/sec  
total size is 21,332,071 speedup is 298.74  
[navy@vm-1-master ~]$
```



- Example of output files

P>XMS{Afk  
P>X~3\$#v  
P>YPM^PNLx  
P>Yig(i;"Zu"  
P>ZJqai<W  
P>%0- T\$P  
P>V-XmJg  
P>lzPQ/K?  
P>+ SZCa  
P>:-|c-Zt  
P>?~vd/nl  
P>\_Mj\_lwBaT  
P>-ZZRBA  
P>-r=s#=#  
P>a2=ml2TL  
P>b0 w8a>  
P>#(ahyNu)  
P>CEfj,p.n  
P>d;:chrL  
P>dwJ3n(BB  
P>fdvuId)vf  
P>glil<-cfR  
P>hl4\$%,"  
P>i+IHU+>  
P>:i=x l'x  
P>=>nqQ) c'  
P>#o~'h%#  
P>#o~,RBcSj  
P>oS~ ^c'i3  
P>o'lBM Z'  
P>Pktlj;s)  
P>qQ'ZM+fr  
P>qS\_l.vmy  
P>rQ9~chD  
P>'r'Rkz->  
P>rchl\_Kly  
P>=SMHlnO?  
P>Sj+SSA(i)  
P>u>|vbVr|  
P>UGWl^H3  
P>un5\_n,%  
P>v/>st<qzg  
P>v>S>60Y  
P>x;:M%Y5L  
P>>xaxyIJo  
P>xSYvwY,\$  
P>y2/Fy4W<  
P>v=FVL'F  
P>vgp3,<6V  
P>[7b]ufJ9  
P>?Sjl.i.to  
P>[9+p].>#  
P>[MD5Fv3]  
P>[Uz/(H/)<  
P>[PkB.d14  
P>[-j-rxu0  
P?=-9Ju>J  
P??|zAjiv  
P?#N4wh7z  
P?#strT7+a  
P?S&~#eNs  
P?sp>bWXF  
P?,ZO#Kg0  
P?SfFLD2x  
P?A%As:,J1  
P?[ldya]0  
P?(.IT8M,\_  
P?[@c(fPa  
P?>=biToK  
P?+=MR2T&5  
P?>7J1J>6  
P?>Bp1Ctq3  
P?,JDOD)26  
P?>2pDc);6  
P?>R rPUne  
P?0 I)+LF<  
P?0 >EDCag  
P?0Xn<j.SL  
P?0bkN6h<  
P?0v~ -yEs  
P?1#M:=JMY  
P?1CLIJAY

0000000000000000000000000564BES90  
000000000000000000000000000864SA515  
0000000000000000000000000008D96670B  
000000000000000000000000000893331  
0000000000000000000000000005649352A  
0000000000000000000000000008798E4  
00000000000000000000000000038543EF9  
00000000000000000000000000086476BD3  
000000000000000000000000000864578A9  
0000000000000000000000000003853036F  
0000000000000000000000000008641BF5  
0000000000000000000000000008D96136E  
0000000000000000000000000008D9341BA  
0000000000000000000000000564C098B  
0000000000000000000000000008D95E72C  
00000000000000000000000000038535AFE  
000000000000000000000000000564CB387  
0000000000000000000000000003850B3C0  
00000000000000000000000000086467129  
0000000000000000000000000008D95D0BE  
00000000000000000000000000038529D05  
0000000000000000000000000008D954078  
00000000000000000000000000038525858  
00000000000000000000000005640DB28  
0000000000000000000000000008D95FB82  
0000000000000000000000000564D3BBE  
0000000000000000000000000564AE0D2  
0000000000000000000000000008D937364  
00000000000000000000000000086456412  
0000000000000000000000000564B0AF4  
0000000000000000000000000008D959329  
00000000000000000000000000038532625  
0000000000000000000000000008D96D136  
00000000000000000000000000038514EFB  
0000000000000000000000000003851EFFB  
000000000000000000000000056497EA9  
0000000000000000000000000008647C98  
0000000000000000000000000008648445E  
0000000000000000000000000003853889F  
0000000000000000000000000008645E942  
00000000000000000000000000086479D49  
00000000000000000000000000038509775  
00000000000000000000000000086461168  
000000000000000000000000000385317FC  
0000000000000000000000000008D9770AT  
00000000000000000000000000086490AOC  
0000000000000000000000000008644D238  
0000000000000000000000000008D940D8  
0000000000000000000000000564BS4DC  
000000000000000000000000056497E09  
000000000000000000000000000385017FA  
0000000000000000000000000008D96A3CF  
0000000000000000000000000008D97AC9C  
0000000000000000000000000564EEC58  
0000000000000000000000000564BEE19  
0000000000000000000000000008D9411C6  
000000000000000000000000000385E20F  
0000000000000000000000000564CA336  
0000000000000000000000000008D978459  
0000000000000000000000000003850724F  
0000000000000000000000000008D94FE67  
0000000000000000000000000008D93FO52  
0000000000000000000000000003851D9D4  
0000000000000000000000000008646A1C2  
00000000000000000000000000038519A51  
0000000000000000000000000008D979224  
0000000000000000000000000564C9075  
000000000000000000000000000834ADFEB2  
0000000000000000000000000008D95D213  
0000000000000000000000000008646E19F  
0000000000000000000000000008D94C8A9  
000000000000000000000000000385C018  
00000000000000000000000000038503A30  
0000000000000000000000000564A02EB  
00000000000000000000000000038538B86  
0000000000000000000000000564BD6DB  
000000000000000000000000000384FB2DE  
0000000000000000000000000564BB824  
0000000000000000000000000564BC81B  
000000000000000000000000056483794  
0000000000000000000000000008D953EC6  
000000000000000000000000000384FAC86  
4444BBBBDDDBBBDDDD888822226666FFFAAA222233331111  
9999DDDD8888CCC22224444888855553333DDDD000077772222  
44445555AAAA6666000082224444444444443333CCCC44444444  
BBBBB6666111155550000CCCAA2222228888CCC888899996666  
6666555555555555CC5555999911117777BBBBFFFF0000DDDD7777  
CCCC2222EEEE0000CCC55557777BBBB0000EEEEBBBB2222DDDD  
DDDDFFFFAAAABBBB

# Result

- But we have some problems
- Sometimes it preserves the total size of inputs
- Sometimes it loses some bytes
- Why?

# What we learned

- The importance of communication
- The importance of the early design <- we cannot resolve the file size issue